

Unit-2 Cryptographic Algorithms

- 1. Data Encryption Standard (DES)**
- 2. Advanced Encryption Standard (AES)**
- 3. Diffie-Hellman key exchange**
- 4. RSA algorithm**
- 5. Basic Concepts of Fields: Groups, Rings, Fields, Modular Arithmetic, Galois Fields,**

Data Encryption Standard (DES)

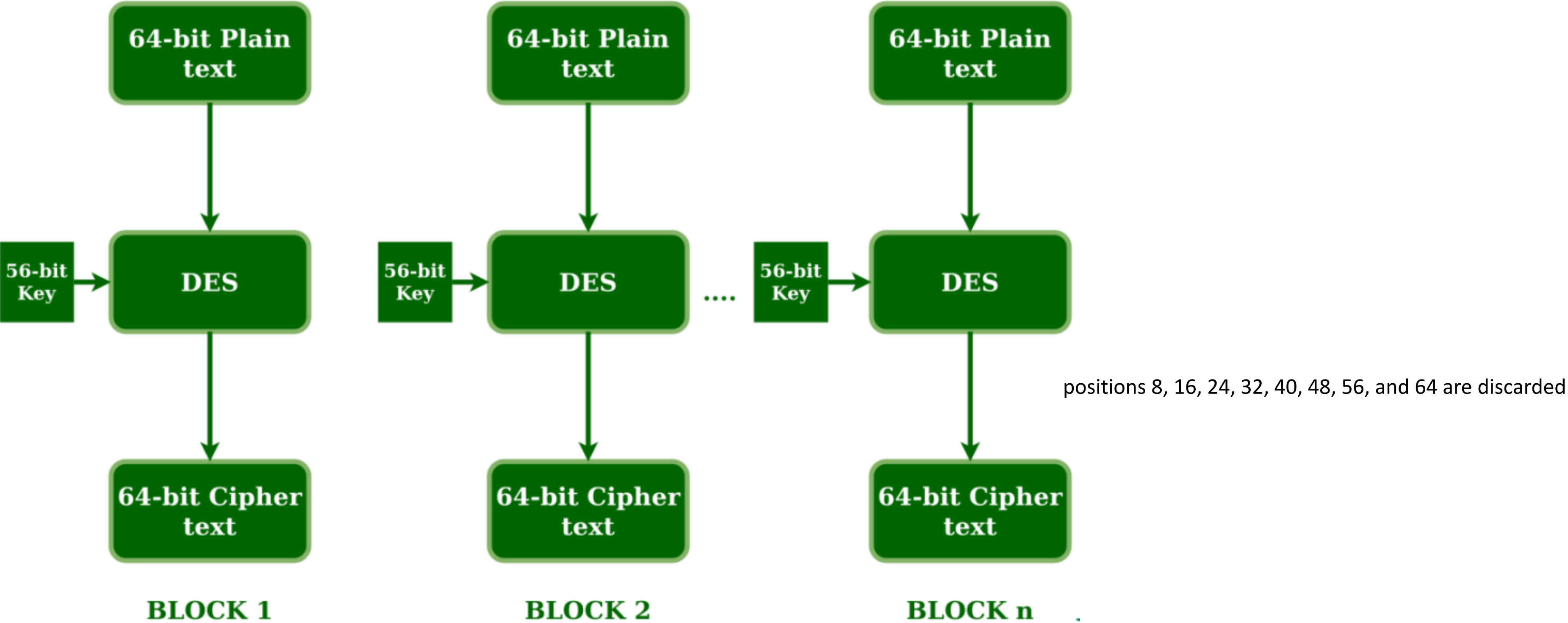
- Data Encryption Standard (DES) is a **symmetric key block cipher algorithm** that was developed by **IBM in the early 1970s** and **adopted as a federal standard in the United States in 1977**.
- The Data Encryption Standard (DES) is a **symmetric-key block cipher published by the National Institute of Standards and Technology (NIST)**.
- DES uses a **56-bit key to encrypt** data in **blocks of 64 bits**. It is a widely used encryption algorithm for **securing sensitive data in various applications, such as financial transactions, e-commerce, and government communications**.
- The DES encryption process involves several stages, including **initial permutation, 16 rounds of substitution and permutation, and final permutation**.
- Each round consists of a **substitution step**, where **bits of the input block are replaced with new bits based on the key**, and a **permutation step**, where the **order of the bits is changed**.
- While DES was considered highly secure when it was **first introduced, advances in computing technology** over the years have made it vulnerable to attacks. As a result, its use has been largely replaced by **newer encryption standards**, such as **Advanced Encryption Standard (AES)**, which provide **stronger** security and **better** performance.
- Nonetheless, DES is still used in some legacy systems and for backward compatibility purposes.

Data Encryption Standard (DES)

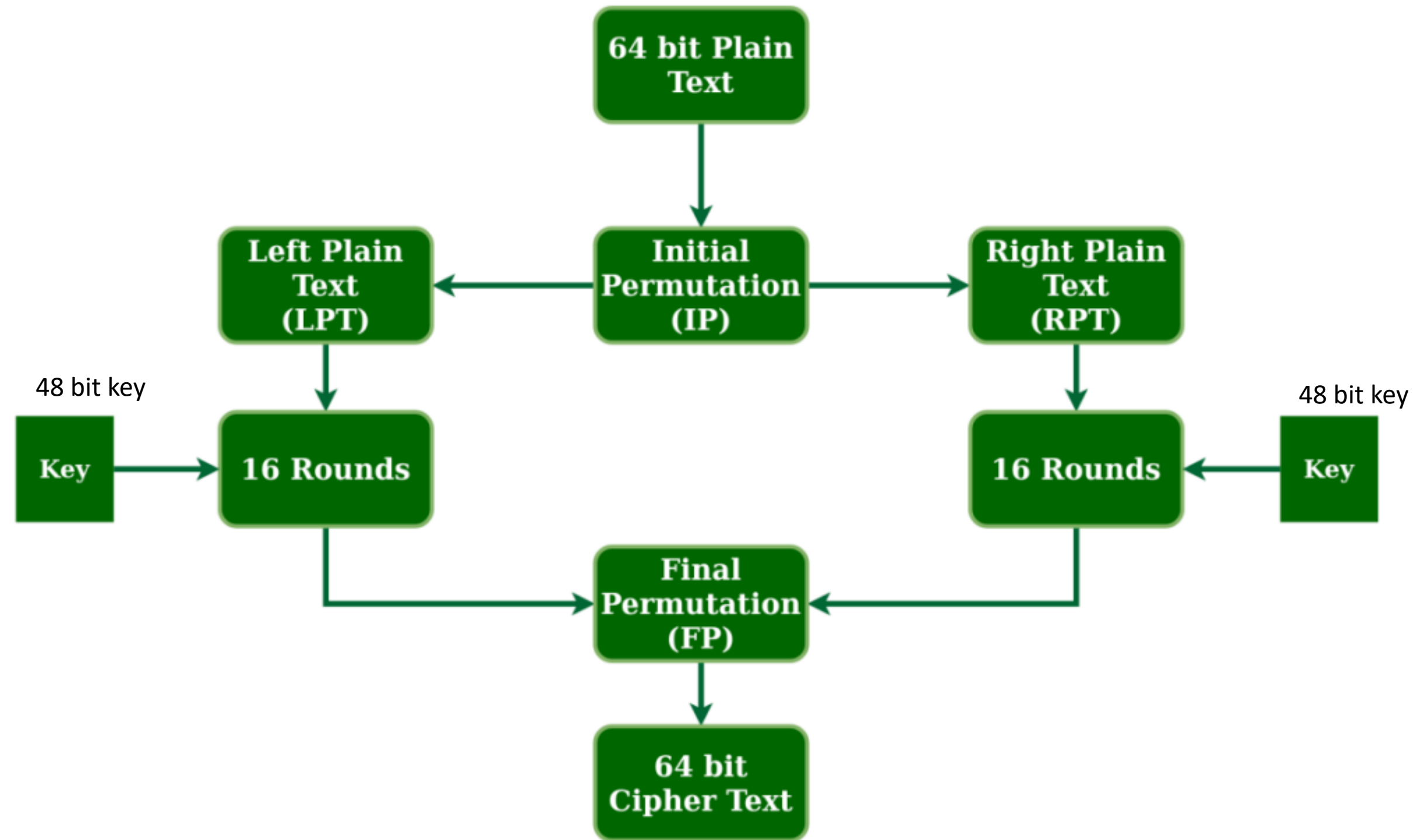
Data Encryption Standard (DES) involves the following steps:

1. **Key Generation:** A 64-bit secret key is generated by the encryption system. This key is used to encrypt and decrypt data.
2. **Initial Permutation:** The 64-bit plaintext is permuted according to a fixed table to produce a new 64-bit block.
3. **Substitution and Permutation:** The plaintext is divided into **16 blocks of 32 bits each**. Each block is then encrypted using **a key generated from the secret key**. This encryption process involves several steps, including **substitution of bits and permutation of the bits** based on the key.
4. **Final Permutation:** The 16 blocks of ciphertext are **combined into a single 64-bit block**, which is then permuted according to a fixed table to produce the final ciphertext.

Data Encryption Standard (DES)



Data Encryption Standard (DES)



Broad Level Steps in DES

Data Encryption Standard (DES)

➤ The decryption process of DES (Data Encryption Standard) is essentially the **reverse of the encryption process**. To decrypt the encrypted message, the receiver uses the **same key that was used by the sender for encryption**. The decryption process involves the following steps:

1. The receiver inputs the ciphertext (encrypted message) into the DES algorithm.
2. The DES algorithm applies **a series of 16 rounds of permutation and substitution** to the ciphertext using the **same key** that was used for encryption.
3. In each round, the DES algorithm **uses a 48-bit subkey** that is derived from the original 64-bit key.
4. The ciphertext is **divided into 64-bit blocks**, and each block is processed separately.
5. In the final round of decryption, the output of the **16th round is fed into an inverse permutation** to obtain the plaintext (original message).
6. The plaintext message is then retrieved and read by the receiver.

Data Encryption Standard (DES)

- DES (Data Encryption Standard) is an encryption algorithm that was widely used in the 1970s and 1980s. While it was a popular encryption method at the time, it is now considered outdated and insecure. Here are some of the disadvantages of DES:
- 1. **Key length:** DES has a key length of only 56 bits, which is considered insufficient for modern encryption requirements. With advances in computing power, it is now possible to crack a 56-bit key in a matter of hours using brute force attacks.
- 2. **Vulnerability to attacks:** DES has several known vulnerabilities that can be exploited by attackers. For example, it is susceptible to brute force attacks, differential cryptanalysis, and linear cryptanalysis.
- 3. **Limited block size:** DES has a block size of only 64 bits, which means that it can only encrypt data in 64-bit blocks. This limitation can make it less efficient for encrypting large volumes of data.
- 4. **Government involvement:** DES was developed by the US government and was widely used for many years by government agencies. However, there are concerns that the government may have built in backdoors or other weaknesses that could be exploited by attackers.
- 5. **Obsolescence:** DES is an old encryption standard that has been superseded by newer, more secure algorithms like AES (Advanced Encryption Standard). As a result, it is no longer considered a secure encryption method and should not be used for sensitive data.

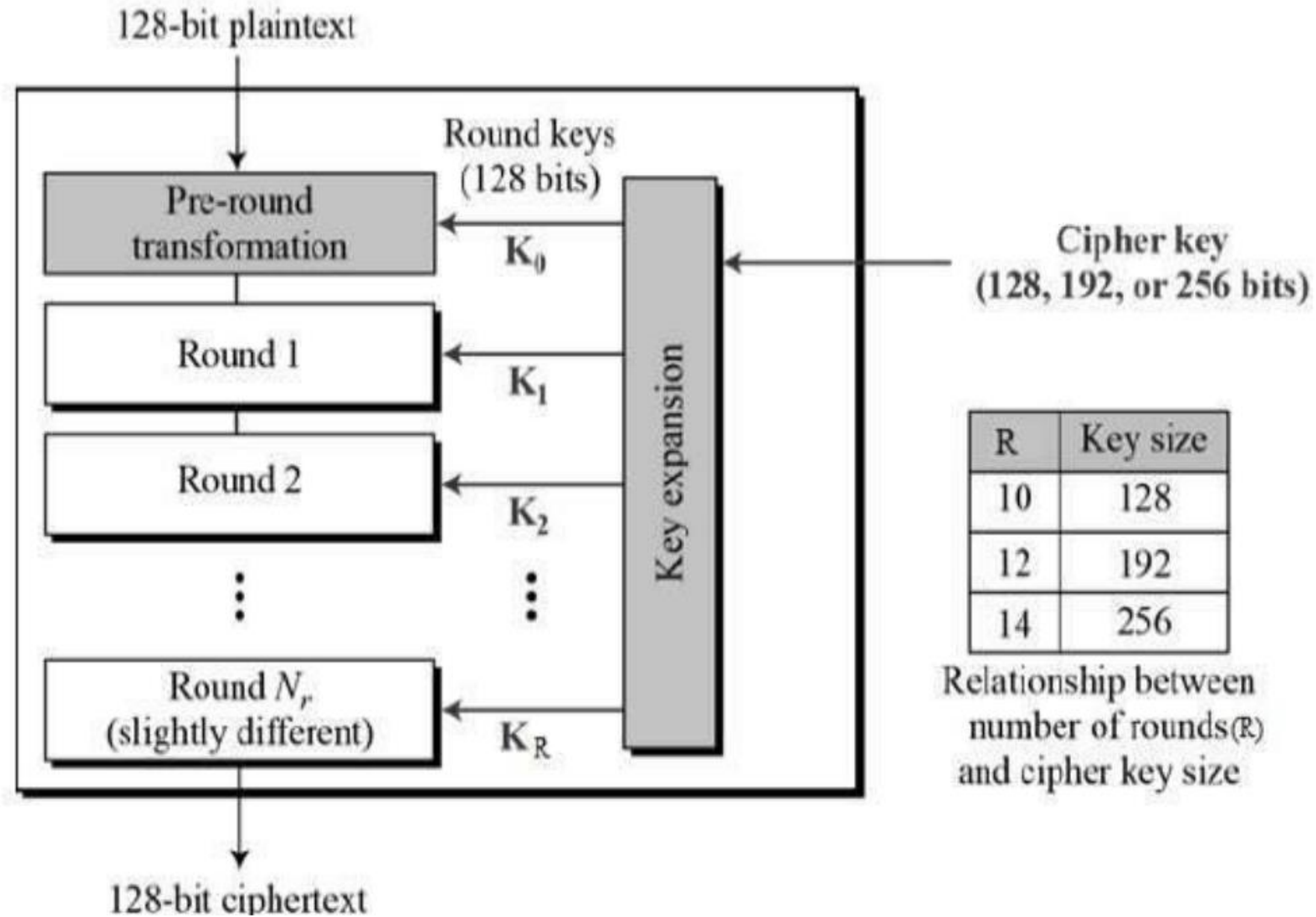
Advanced Encryption Standard (AES)

- Advanced Encryption Standard (AES) is a **symmetric-key encryption algorithm** used to encrypt and decrypt data. It is a widely-used cryptographic algorithm that was selected by the National Institute of Standards and Technology (NIST) in **2001** to **replace the older Data Encryption Standard (DES)**. AES is a block cipher encryption algorithm that encrypts data in **fixed-size blocks of 128 bits** using a key of **128, 192, or 256 bits**.

The encryption process of AES involves the following steps:

1. **Key expansion:** The original key is expanded into a **series of subkeys** that will be used in each round of encryption.
2. **Initial round:** The plaintext is **XORed with the first subkey**.
3. **Rounds:** The **plaintext and the subkey** are passed through a **series of substitution and permutation operations**, known as the **SubBytes and ShiftRows** steps, respectively. These steps are repeated for a certain number of rounds, depending on the key size (**10 rounds for 128-bit keys, 12 rounds for 192-bit keys, and 14 rounds for 256-bit keys**).
4. **Final round:** The plaintext is passed through **one final round of substitution and permutation**, known as the **MixColumns step**, and **XORed with the last subkey**.
5. The resulting ciphertext is then transmitted securely to the recipient.

Advanced Encryption Standard (AES)



Advanced Encryption Standard (AES)

Decryption Process:

1. The decryption process of AES is essentially the **reverse of the encryption process**. The recipient uses the **same key** to decrypt the ciphertext and retrieve the original plaintext. The decryption process involves the following steps:
2. The recipient inputs the **ciphertext** (encrypted message) into the **AES algorithm**.
3. The AES algorithm applies a **series of rounds of substitution and permutation to the ciphertext** using the **same key** that was used for encryption.
4. In each round, the AES algorithm uses a **subkey** that is derived from the **original key**.
5. The **ciphertext** is **divided into 128-bit blocks**, and each block is processed separately.
6. In the final round of decryption, the output of the final **round** is **XORed** with the **last subkey** to obtain the plaintext (original message).
7. The plaintext message is then retrieved and read by the recipient.

Advanced Encryption Standard (AES)

Limitation:

- One of the main limitations of AES is that it is a symmetric encryption algorithm, which means that the **same key is used for encryption and decryption**.
- This makes it vulnerable to attacks in which an attacker **gains access to the key**. Additionally, like all encryption algorithms, AES is only as secure as the key used to encrypt the data. If the **key is weak or easily guessable, an attacker** may be able to easily decrypt the data.

AES Analysis

- In present day cryptography, AES is widely adopted and supported in both hardware and software. Till date, **no practical cryptanalytic attacks against AES** has been discovered. Additionally, AES has built-in flexibility of key length, which allows a degree of '**future-proofing**' against progress in the ability to perform exhaustive key searches.
- However, just as for DES, the AES security is assured only if it is correctly implemented and **good key management** is employed.

Diffie-Hellman key exchange

Diffie-Hellman key exchange is a cryptographic protocol used to establish a shared **secret key** between two parties over an insecure communication channel. This key can then be used to encrypt subsequent communication between the two parties.

1. The process of the Diffie-Hellman key exchange involves the following steps:
2. The two parties, let's call them **Sita** and **Ram**, agree on a **public prime number p** and a **public base g**.
3. **Sita** chooses a secret number a and calculates $A = g^a \bmod p$.
4. **Ram** chooses a secret number b and calculates $B = g^b \bmod p$.
5. **Sita** and **Ram** exchange A and B with each other.
6. **Sita** calculates the shared secret key as $K = B^a \bmod p$.
7. **Ram** calculates the shared secret key as $K = A^b \bmod p$.

Diffie-Hellman

Sita and Ram agree on a public prime number $p = 23$ and a public base $g = 5$.

Sita chooses a secret number $a = 6$ and calculates $A = g^a \bmod p = 5^6 \bmod 23 = 8$.

Ram chooses a secret number $b = 15$ and calculates $B = g^b \bmod p = 5^{15} \bmod 23 = 19$.

Sita and Ram exchange $A = 8$ and $B = 19$ with each other.

Sita calculates the shared secret key as $K = B^a \bmod p = 19^6 \bmod 23 = 2$.

Ram calculates the shared secret key as $K = A^b \bmod p = 8^{15} \bmod 23 = 2$.

So the shared secret key that Sita and Ram have **established is $K = 2$** . They can now use this key to encrypt their subsequent communication

Diffie-Hellman

- The limitation of the Diffie-Hellman key exchange is that it **does not provide authentication**. It only establishes **a shared secret key** between two parties, but it does not verify the identity of those parties. As a result, an attacker could potentially **impersonate one of the parties** and intercept the communication.
- The application of Diffie-Hellman key exchange is primarily in **secure communication protocols** such as SSL/TLS, SSH, and VPNs.
- These protocols use the Diffie-Hellman key exchange to establish a shared secret key between the client and server, which is then used for **subsequent encrypted communication**.

Diffie-Hellman

- The RSA algorithm is a widely used **public-key cryptography** algorithm that was invented by **Ron Rivest, Adi Shamir, and Leonard Adleman** in 1977. It is named after the initials of their last names.
- The RSA algorithm is **based on** the fact that it is easy **to multiply two large prime numbers** together, but it is **extremely difficult to factor the product back into its constituent primes**.
- This property is used to generate two large prime numbers, **p and q, which are used as the private key for encryption and decryption**.

The RSA algorithm works as follows:

Key generation:

1. Choose two large prime numbers, p and q .
2. Calculate $n = p * q$ and the totient¹ of n , $\phi(n) = (p-1) * (q-1)$.
3. Choose a **public exponent** e such that $1 < e < \phi(n)$ and² **gcd**($e, \phi(n)$)=1.
4. Calculate the **private exponent** d such that d is the multiplicative inverse of e modulo $\phi(n)$, i.e., $d * e \equiv 1 \pmod{\phi(n)}$ ³i.e. $d=(1 +k * \phi(n))/e$
5. The public key is **(n, e)** and the private key is **(n, d)**.

¹ It is the number of integers k in the range $1 \leq k \leq n$ for which the greatest common divisor $\gcd(n, k)$ is equal to 1

² Greatest Common Divisor (GCD))

³ Using extended Euclidean algorithm

The RSA algorithm works as follows:

Encryption:

1. The sender encrypts a message M by computing $C = M^e \bmod n$ using the **public key** (n, e) .
2. The sender then sends the encrypted message C to the recipient.

Decryption:

1. The recipient decrypts the message C by computing $M = C^d \bmod n$ using the **private key** (n, d) .

Note: $M < N$

The RSA algorithm works as follows:

Key generation:

1. Choose two prime numbers, $p = 11$ and $q = 3$.
2. Compute $n = p * q = 33$ and $\phi(n) = (p-1) * (q-1) = 20$.
3. Choose a public exponent e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$. Let's choose $e = 7$.
4. Compute the private exponent d such that $d * e \equiv 1 \pmod{\phi(n)}$. One way to compute d is to use the **extended Euclidean algorithm** $d = (1 + k * \phi(n)) / e$. We find that $d = 3$.
5. The **public key** is $(n, e) = (33, 7)$ and the **private key** is $(n, d) = (33, 3)$.

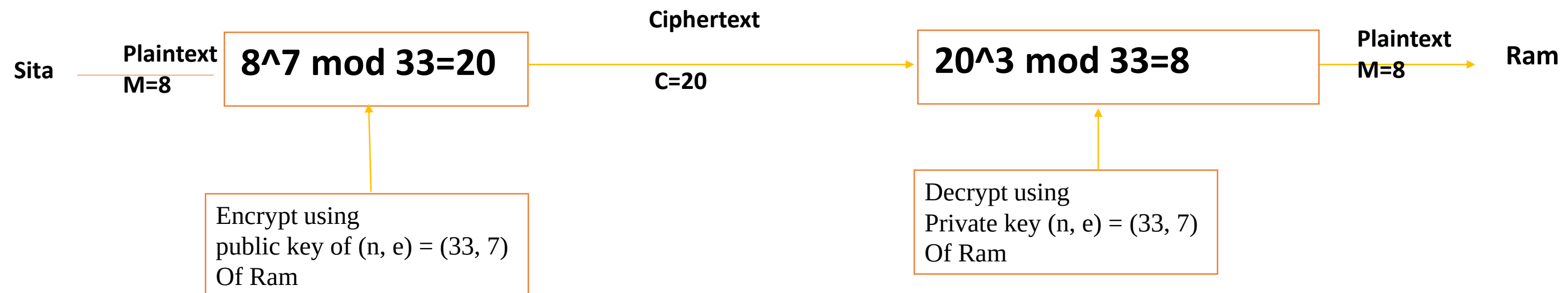
The RSA algorithm:

Encryption:

1. Suppose **Sita** wants to send a message $M = 8$ to **Ram**. Sita encrypts the message by computing $C = M^e \bmod n = 8^7 \bmod 33 = 20$.
2. **Sita** sends the encrypted message $C = 20$ to **Ram**.

Decryption:

1. **Ram** receives the encrypted message $C = 20$. He decrypts it by computing $M = C^d \bmod n = 20^3 \bmod 33 = 8$.
2. **Ram** has successfully decrypted the message, which is 8.



The RSA algorithm:

1. Perform encryption and decryption using RSA algorithms where $p=3$, $q=11$, $e=?$ $M=5$, $d=?$
2. Perform encryption and decryption using RSA algorithms where $p=3$, $q=5$, $e=?$ $M=8$, $d=?$

The RSA algorithm:

Application:

1. The RSA algorithm has many applications, including **secure communication, digital signatures, and key exchange**.
2. It is widely used in **internet protocols such as SSL/TLS, SSH, and HTTPS**.

Limitation:

1. One limitation of the RSA algorithm is that it is **computationally expensive, particularly for large keys**.
2. Another limitation is that it is vulnerable to certain attacks, such as the **timing attack and the side-channel attack**, which exploit weaknesses in the implementation of the algorithm rather than the algorithm itself.
3. To mitigate these limitations, it is important to use secure implementations and **key sizes that are appropriate** for the level of security required.

A Group

- **A group is a mathematical structure** consisting of a **set of elements** and a **binary operation that combines any two elements to form a third element**, with the **operation satisfying certain axioms**. Here's an example of a group:
- Consider the set of integers under addition. That is, the set $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ and the operation of addition. This **set and operation form a group**. Here are the four group axioms that are satisfied:
 - 1. Closure:** For any two integers a and b , $a + b$ is also an integer. So, the set of integers is closed under addition.
 - 2. Associativity:** For any integers a , b , and c , $(a + b) + c = a + (b + c)$.
 - 3. Identity:** There exists an integer 0 such that for any integer a , $a + 0 = 0 + a = a$. In this case, the identity element is 0 .
 - 4. Inverse:** For any integer a , there exists an integer $-a$ such **that $a + (-a) = (-a) + a = 0$** . In other words, every element in the group has an inverse. For example, the inverse of 3 is -3 , and the inverse of -5 is 5 .
- Therefore, the set of integers under addition satisfies the **four group axioms and is a group**.
- *Note that there are many other examples of groups, such as the group of invertible matrices under matrix multiplication, or the group of permutations of a finite set under function composition.*

Ring

In mathematics, rings are a **type of algebraic structure** that consists of a **set of elements together with two binary operations called addition and multiplication**, which satisfy certain axioms. Here are a few examples of rings:

1. **The integers** (denoted by the symbol $\mathbb{Z}$) form a ring, **with addition and multiplication as the binary operations**. The integers satisfy the ring axioms, such as the **associative** and **distributive** properties.

Associative property of addition: For any elements a , b , and c in a ring R , we have:

$$(a + b) + c = a + (b + c)$$

For example, in the ring of integers $\mathbb{Z}$, we have:

$$(2 + 3) + 4 = 5 + 4 = 9$$

$$2 + (3 + 4) = 2 + 7 = 9$$

These two **expressions have the same value**, which shows that addition is associative in the ring of integers.

Distributive property of multiplication: For any elements a , b , and c in a ring R , we have:

$$a * (b + c) = (a * b) + (a * c)$$

For example, in the **ring of polynomials with coefficients** in a field $F[x]$, we have:

$$x * (2x^2 + 3x + 4) = 2x^3 + 3x^2 + 4x$$

$$x * 2x^2 + x * 3x + x * 4 = 2x^3 + 3x^2 + 4x$$

2. **The polynomials with coefficients** in a field (such as the real numbers or complex numbers) form a ring, denoted by the symbol $F[x]$, where F is the field. Addition and multiplication of polynomials are defined in the usual way, and they satisfy the ring axioms.

3. **The matrix algebra** over a field forms a ring, denoted by the symbol $M(n, F)$, where n is a positive integer and F is a field. Addition and multiplication of matrices are defined in the usual way, and they satisfy the ring axioms.

4. **The quaternions form a non-commutative ring**, denoted by the symbol H . The quaternions are a four-dimensional algebra over the real numbers, with multiplication defined in a non-commutative way

Field

In mathematics, a field is a **set of elements together with two binary operations called addition and multiplication**, which satisfy certain axioms. The main **difference between a field and a ring is that in a field, every nonzero element has a multiplicative inverse**. Here are a few examples of fields:

1. **The real numbers** (denoted by the symbol $\mathbb{R}$) form a field, with addition and multiplication as the binary operations. Every nonzero real number has a multiplicative inverse (for example, the **multiplicative inverse of 2** is $1/2$).
2. **The complex numbers** (denoted by the symbol $\mathbb{C}$) form a field, with addition and multiplication as the binary operations. Every nonzero complex number has a multiplicative inverse (for example, the multiplicative inverse of $1 + i$ is $(1 - i)/2$).
3. **The rational numbers** (denoted by the symbol $\mathbb{Q}$) form a field, with addition and multiplication as the binary operations. Every nonzero rational number has a multiplicative inverse (for example, the multiplicative inverse of $2/3$ is $3/2$).
4. **The field of p-adic numbers** (denoted by the symbol $\mathbb{Q}_p$) is a field that extends the rational numbers $\mathbb{Q}$, and is used in number theory and algebraic geometry. The p-adic numbers are constructed by completing the rational numbers with respect to a certain non-Archimedean norm.

Modular Arithmetic

- Modular arithmetic is a type of arithmetic that deals with **integers modulo a fixed integer** called the modulus. Here is an example of modular arithmetic:
- Suppose we are working modulo 7. This means that we are only considering remainders when integers are divided by 7. For example, the remainders of 10, 14, and 21 when divided by 7 are 3, 0, and 0, respectively.
- Addition in modular arithmetic works by adding the remainders modulo 7. For example, to calculate $3 + 4$ modulo 7, we first add 3 and 4 to get 7, and then take the remainder when 7 is divided by 7, which is 0. Therefore, $3 + 4$ is equivalent to 0 modulo 7.
- Multiplication in modular arithmetic works by multiplying the remainders modulo 7. For example, to calculate $3 * 4$ modulo 7, we first multiply 3 and 4 to get 12, and then take the remainder when 12 is divided by 7, which is 5. Therefore, $3 * 4$ is equivalent to 5 modulo 7.
- Modular arithmetic is often used in number theory and cryptography, where it provides a useful tool for solving problems and creating secure codes.

Galois Fields

Galois fields, also known as **finite fields**, are fields with a **finite number of elements**. The number of elements in a Galois field is called **the order of the field**, and is denoted by q , where **q is a prime power** (i.e., $q = p^k$ for some prime number p and positive integer k).

Here is an example of a Galois field:

Consider the Galois field $GF(5^2)$, which has order $5^2 = 25$. Elements of this field can be represented as polynomials of degree 1 or less with coefficients in $GF(5)$ (i.e., the field with 5 elements). For example, one possible element of $GF(5^2)$ is $2x + 3$, where x is an indeterminate and the coefficients 2 and 3 belong to $GF(5)$.

Addition and multiplication in $GF(5^2)$ are defined by adding and multiplying these polynomials modulo a irreducible polynomial of degree 2 with coefficients in $GF(5)$. One possible choice for this irreducible polynomial is $x^2 + 2x + 4$.

For example, to add two elements of $GF(5^2)$, we add their coefficients modulo 5 and reduce the result modulo $x^2 + 2x + 4$. For instance:

$$(2x + 3) + (4x + 1) = 6x + 4 = x + 4 \pmod{x^2 + 2x + 4}$$

To multiply two elements of $GF(5^2)$, we use the distributive property of polynomials and reduce the result modulo $x^2 + 2x + 4$. For example:

$$(2x + 3) * (4x + 1) = 8x^2 + 14x + 3 = 3x^2 + 4x + 3 \pmod{x^2 + 2x + 4}$$

Galois fields have applications in many areas of mathematics and computer science, including coding theory, cryptography, and finite geometry.

Polynomial arithmetic

Polynomial arithmetic is a type of arithmetic that **deals with polynomials**, which are expressions **consisting of a sum of terms**, each of which is a **constant multiple of a variable raised to a non-negative integer power**. Polynomial arithmetic includes addition, subtraction, multiplication, and division of polynomials.

Suppose we want to multiply the polynomials $f(x) = x^2 + 2x + 1$ and $g(x) = x + 3$. To do this, we use the distributive property of multiplication:

$$f(x) * g(x) = (x^2 + 2x + 1) * (x + 3) = x^3 + 3x^2 + 2x^2 + 6x + x + 3 = x^3 + 5x^2 + 7x + 3$$

Now, suppose we want to divide the polynomial $h(x) = x^3 - 4x^2 + 3x - 2$ by the polynomial $p(x) = x - 2$. To do this, we use polynomial long division:

$$\begin{array}{r} x - 2 \overline{) x^3 - 4x^2 + 3x - 2} \\ \underline{-x^3 + 2x^2} \\ -2x^2 + 3x \\ \underline{2x^2 - 4x} \\ 7x - 2 \\ \underline{7x - 14} \\ 12 \end{array}$$

Therefore, the quotient polynomial is $q(x) = x^2 + 2x + 1$, and the remainder polynomial is $r(x) = 12$.